Notes on lecture 8

Jiten Chandra Kalita and Shyamashree Upadhyay

1 Maximum and minimum values of a function of 2 variables

Before we start this topic, we need a notation, which we state first:

Notation 1.0.1. Let (a, b) be a point in $\mathbb{R}^2$ and let $\delta > 0$ be a positive real number. Let us denote by $D_\delta(a, b)$ a disk centered at (a, b) of radius δ. □

Let us now move on to the topic of maximum and minimum values of a function of 2 variables.

Definition 1.0.2. Let f be a function of 2 variables x and y. Let (a, b) be a point in the domain of definition of f. Then f is said to have a **local maximum** at (a, b) if $f(x, y) \leq f(a, b)$ for all $(x, y) \in D_\delta(a, b)$ for some $\delta > 0$ such that $D_\delta(a, b)$ is contained in the domain of definition of f. The number $f(a, b)$ is called a **local maximum value**.
Likewise, the function f is said to have a **local minimum** at (a, b) if $f(x, y) \geq f(a, b)$ for all $(x, y) \in D_\delta(a, b)$ for some $\delta > 0$ such that $D_\delta(a, b)$ is contained in the domain of definition of f. The number $f(a, b)$ is called a **local minimum value**. □

Definition 1.0.3. Suppose we consider the domain of f and in that domain we consider all the local maxima and the local minima. The largest of all the local maxima is called the **absolute maximum** and the smallest of all the local minima is called the **absolute minimum**. □

Figure 1.0.1 explains Definitions 1.0.2 and 1.0.3.

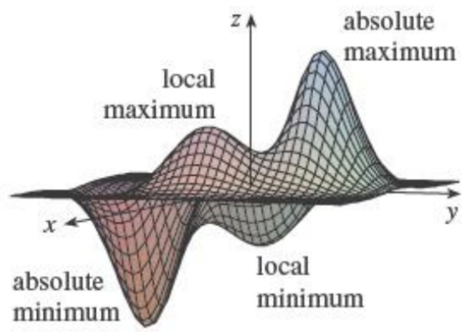


Figure 1.0.1: Maximum and minimum of a function

Theorem 1.0.4. *If f has a local maximum or minimum at (a, b) and the first order partial derivatives of f exist there, then $f_x(a, b) = f_y(a, b) = 0$.*

PROOF: Suppose f has a local maximum or minimum at (a, b). Let

$$g(x) = f(x, b).$$

If $z = f(x, y)$ is the surface representing f, then we are simply considering the curve of intersection of this surface with the plane $y = b$. Then

$$g'(a) = \lim_{h\to 0} \frac{g(a+h) - g(a)}{h}$$

$$= \lim_{h\to 0} \frac{f(a+h, b) - f(a, b)}{h} = f_x(a, b).$$

Now, the existence of maxima or minima for $g(x)$ at the point a is related to the existence of maxima or minima of $f(x, b)$ at the point (a, b). Also, from single variable calculus, we know that for the existence of maxima or minima of a function g (of one variable) at a point a, we must have $g'(a) = 0$. That is, we must have $f_x(a, b) = 0$. Similarly, we must also have $f_y(a, b) = 0$. □

Remark 1.0.5. In the statement of Theorem 1.0.4, it is important that the first order partial derivatives of f exist at (a, b). It may happen that f has a local maximum or minimum at (a, b) but the first order partial derivatives do not exist at (a, b).

Example 1.0.6 explains Remark 1.0.5.

Example 1.0.6. Consider the cone $z = \sqrt{x^2+y^2}$. If we take $f(x,y) = \sqrt{x^2+y^2}$, then clearly f has a local minimum at $(0,0)$ (obvious from the graph of f). But the first order partial derivatives of f do not exist at $(0,0)$ because the function f is not smooth at the origin. $\square$

1.1 Geometrical significance of Theorem 1.0.4

Let $f(x,y)$ have a maximum or minimum at (a,b). Let $c = f(a,b)$. Consider the tangent plane to the surface $z = f(x,y)$ at the point (a,b,c). The equation of this tangent plane is given by

$$z - c = f_x(a,b)(x-a) + f_y(a,b)(y-b).$$

Now since f has a maximum or minimum at (a,b), we must have (by Theorem 1.0.4):

$$f_x(a,b) = f_y(a,b) = 0,$$

provided the partial derivatives f_x and f_y exist at (a,b). Putting $f_x(a,b) = f_y(a,b) = 0$ in the equation of the tangent plane, we get

$$z = c.$$

That is, the equation of the tangent plane becomes $z = c$. But $z = c$ is a plane parallel to the xy-plane. So, at a point of local maximum or minimum, the tangent plane to the surface must be a horizontal plane.
Look at Figure 1.1.1. Observe that at the points marked 1,2,3,4, all the tangent planes to the surface will be horizontal planes. The points 1,2,3,4 are the points of local maxima/minima of the surface in the figure.

2 Critical points

Definition 2.0.1. A point (a,b) is called a **critical point** (or **stationary point**) of f if either $f_x(a,b) = f_y(a,b) = 0$ or one of these (not necessarily both) partial derivatives f_x and f_y does not exist. $\square$

Remark 2.0.2. Theorem 1.0.4 above says that if f has a local maximum or minimum at (a,b), then (a,b) is a critical point of f. However, as in single variable calculus, not all critical points give rise to maxima or minima. At a critical point, a function could have a local maximum or local minimum or neither.

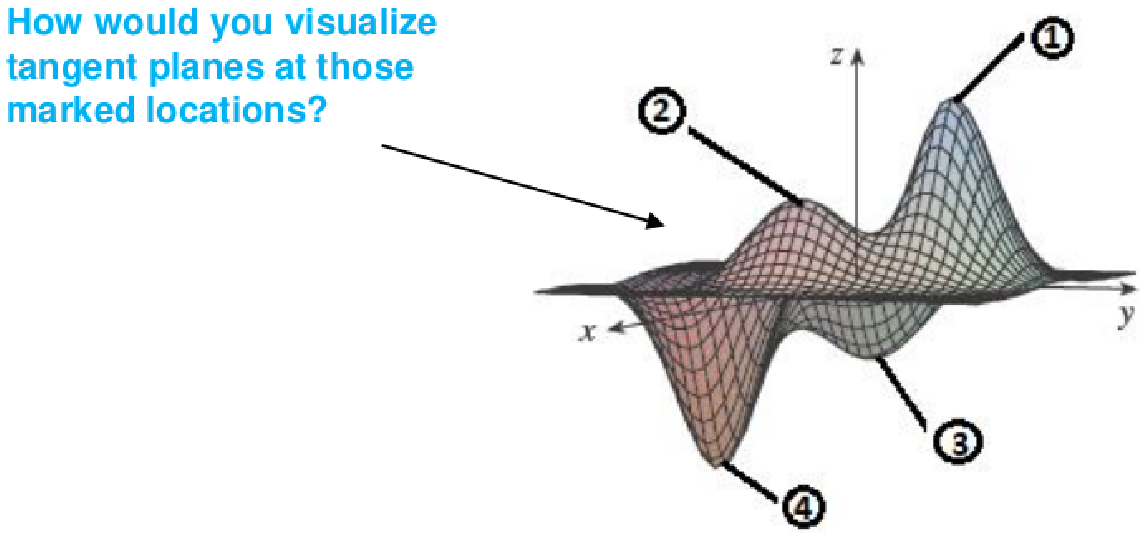


Figure 1.1.1: Tangent planes at local maxima/minima

Example 2.0.3. Let $f(x, y) = x^2 + y^2 - 2x - 6y + 14$.
We can write $f(x, y)$ as $(x - 1)^2 + (y - 3)^2 + 4$. So $z = f(x, y)$ represents a paraboloid. The graph of f is given in Figure 2.0.1. It is clear from the graph of f that f has a minimum at the point $(1, 3)$. If we take the entire $\mathbb{R}^2$ as the domain of f, then it is difficult to say anything about maximum. Here $f_x = 2(x-1)$ and $f_y = 2(y-3)$. So $f_x(1, 3) = f_y(1, 3) = 0$. This means that $(1, 3)$ is a critical point of f. □

The example below shows that a critical point is not necessarily a point of local maximum or minimum.

Example 2.0.4. Let $f(x, y) = x^2 - y^2$. Then

$$f_x = 2x \text{ and } f_y = -2y.$$

Clearly, $(0, 0)$ is a critical point of f. Now if $(0, 0)$ is a point of local maximum, then we must have

$$f(x, y) \leq f(0, 0)$$

for all (x, y) in a neighborhood of $(0, 0)$. Similarly, if $(0, 0)$ is a point of local minimum, then we must have

$$f(x, y) \geq f(0, 0)$$

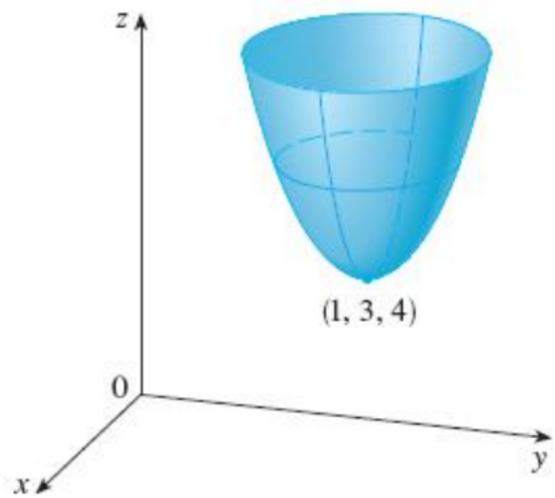


Figure 2.0.1: Figure for Example 2.0.3

for all (x, y) in a neighborhood of $(0, 0)$. But observe that along the x-axis, $f(x, y) = x^2$, and along the y-axis, $f(x, y) = -y^2$. So in any neighborhood of $(0, 0)$, we will get positive values of the f along the x-axis, and negative values of f along the y-axis. So neither

$$f(x, y) \leq f(0, 0)$$

nor

$$f(x, y) \geq f(0, 0)$$

is satisfied in a neighborhood of $(0, 0)$. So f has neither a maximum nor a minimum at $(0, 0)$. □

Remark 2.0.5. The point $(0, 0)$ is a critical point in Example 2.0.4. But it is neither a point of local maximum nor minimum. Such points are known as "**saddle points**", which we will study later.

3 Test for the existence of maxima and minima

The second derivative test: Suppose the second order partial derivatives of f are continuous on a disk with center (a, b), and suppose that $f_x(a, b) = f_y(a, b) = 0$ (that is, (a, b) is a critical point of f). Let

$$D = D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2.$$

(a) If $D > 0$ and $f_{xx}(a, b) > 0$, then $f(a, b)$ is a local minimum.
(b) If $D > 0$ and $f_{xx}(a, b) < 0$, then $f(a, b)$ is a local maximum.
(c) If $D < 0$, then $f(a, b)$ is not a local maximum or minimum.
Note 1: In case (c), the point (a, b) is called a **saddle point** of f and the graph of f crosses its tangent plane at (a, b).
Note 2: If $D = 0$, the second derivative test gives no information. f could have a local maximum or local minimum at (a, b) or (a, b) could be a saddle point of f.
Note 3: To remember the formula for D, it is helpful to write it as a determinant:

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - (f_{xy})^2.$$

Example 3.0.1. Find the local maximum and minimum values and saddle points of $f(x, y) = x^4 + y^4 - 4xy + 1$.
Figure 3.0.1 shows a graph of f. It is easy to calculate that

$$f_x(x, y) = 4x^3 - 4y,\ f_y(x, y) = 4y^3 - 4x,\ f_{xx}(x, y) = 12x^2,$$

$$f_{yy}(x, y) = 12y^2,\ f_{xy}(x, y) = f_{yx}(x, y) = -4.$$

For getting the critical points, we equate f_x and f_y to zero. This gives

$$y = x^3 \text{ and } x = y^3. \qquad (*)$$

From $(*)$, we get $x^9 - x = 0$, or

$$x(x^4 + 1)(x^2 + 1)(x - 1)(x + 1) = 0.$$

We are not bothered about the factors $x^4 + 1$ and $x^2 + 1$ above because they will give some complex roots. So the other factors give

$$x = 0, 1, -1.$$

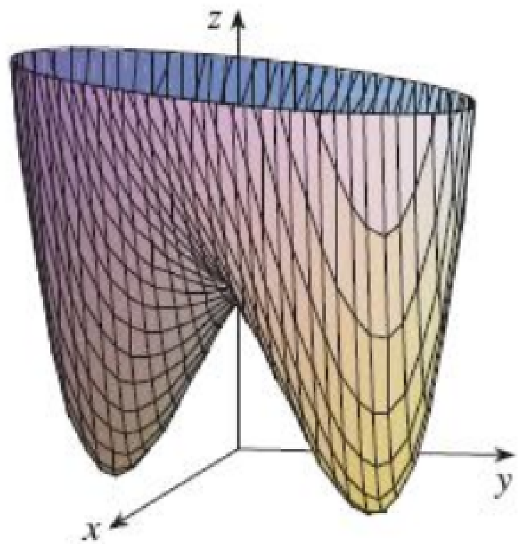


Figure 3.0.1: Figure for Example 3.0.1

Hence

$$y = 0, 1, -1. \text{ (Since } y = x^3\text{)}$$

So the critical points are $(0,0)$, $(1,1)$, and $(-1,-1)$. Again,

$$D(x,y) = f_{xx}f_{yy} - (f_{xy})^2 = 144x^2y^2 - 16.$$

Therefore, $D(0,0) = -16 < 0$. So $(0,0)$ is a saddle point.
Observe that

$$D(1,1) = D(-1,-1) = 128 > 0.$$

So at $(1,1)$ and $(-1,-1)$, there is a possibility of existence of maxima or minima. Now

$$f_{xx}(1,1) = f_{xx}(-1,-1) = 12 > 0.$$

So f has a local minimum at both the points $(1,1)$ and $(-1,-1)$. □

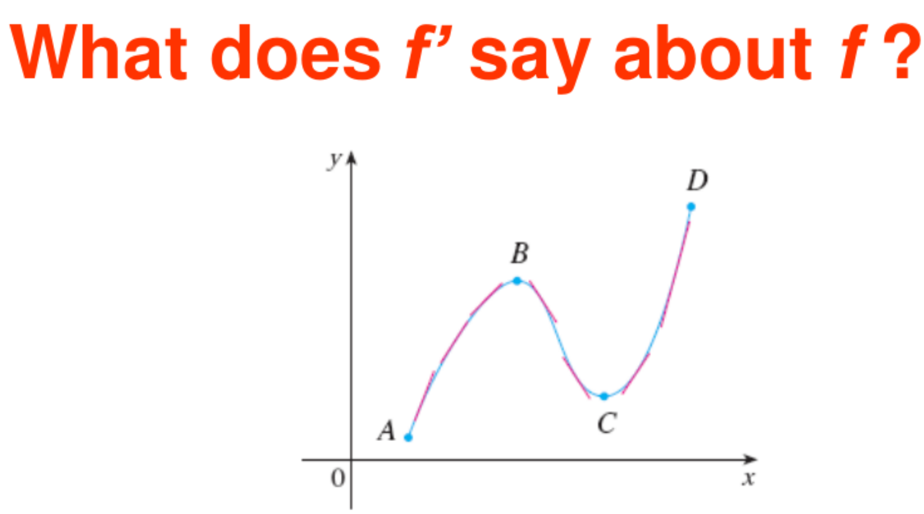
Question: In the statement of the second derivative test, the condition is on f_{xx}. Why not on f_{yy}?
Answer: We know that

$$D(x,y) = f_{xx}f_{yy} - (f_{xy})^2.$$

Now for the existence of local maximum or minimum, we have $D > 0$. Also, in the expression of D, we have $(f_{xy})^2$, which is a non-negative quantity. So this implies that both f_{xx} and f_{yy} must have the same sign (so that their product is always > 0). That's why it is sufficient to put the condition on f_{xx} only.

3.1 Proof of the second derivative test

In this subsection, we will prove part(a) of the second derivative test. But before that, let us see how derivatives affect the shape of a graph of a function (of a single variable). Based on that, we will prove part(a) of the second derivative test.

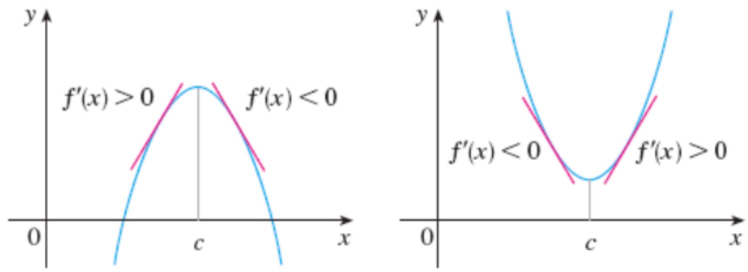


Increasing/Decreasing Test

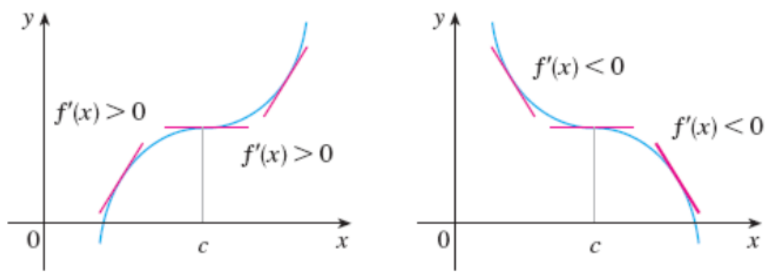
(a) If $f'(x) > 0$ on an interval, then f is increasing on that interval.

(b) If $f'(x) < 0$ on an interval, then f is decreasing on that interval.

The above figure is a graph of a function of one variable. We have drawn some tangents at different points on this graph.



(a) Local maximum (b) Local minimum



(c) No maximum or minimum (d) No maximum or minimum

The First Derivative Test Suppose that c is a critical number of a continuous function f.

(a) If f' changes from positive to negative at c, then f has a local maximum at c.

(b) If f' changes from negative to positive at c, then f has a local minimum at c.

(c) If f' does not change sign at c (for example, if f' is positive on both sides of c or negative on both sides), then f has no local maximum or minimum at c.

Look at Figure 3.1.1 below. Observe that for part (a) in this figure, the slopes of the tangent lines are getting steeper and steeper as we move up. So the slope of the tangent lines is a monotonically increasing function. And for part (b) of the same figure, the slope of the tangent lines is a monotonically decreasing function (For part (b), observe that the slopes of the tangent lines are getting flatter and flatter as we move up). So for part (a), $f'' > 0$ and for part (b), we have $f'' < 0$. Also for part (a), observe that the graph of the function lies above the tangent lines. However, for part (b), the graph of the function lies below the tangent lines.

What does f'' say about f ?

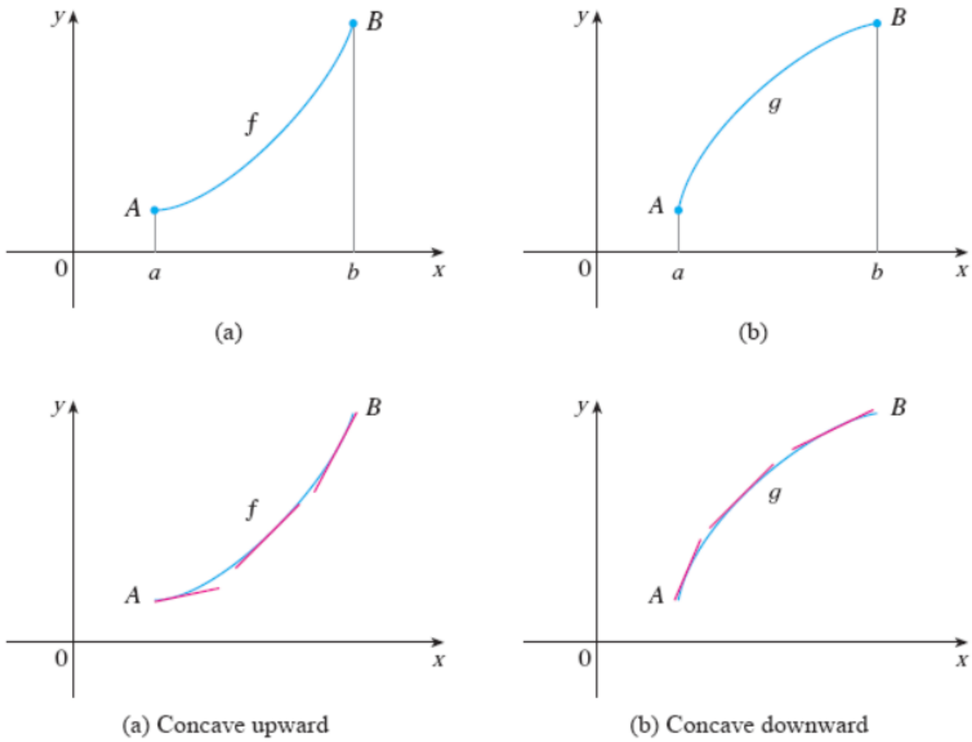


Figure 3.1.1: Concave upward and downward

Definition If the graph of f lies above all of its tangents on an interval I, then it is called **concave upward** on I. If the graph of f lies below all of its tangents on I, it is called **concave downward** on I.

Remark 3.1.1. A shape looking like the english alphabet U means concave upwards, and a shape looking like the inverted english alphabet U means concave downwards.

In Figure 3.1.2, **CU** means concave upwards, and **CD** means concave downwards.

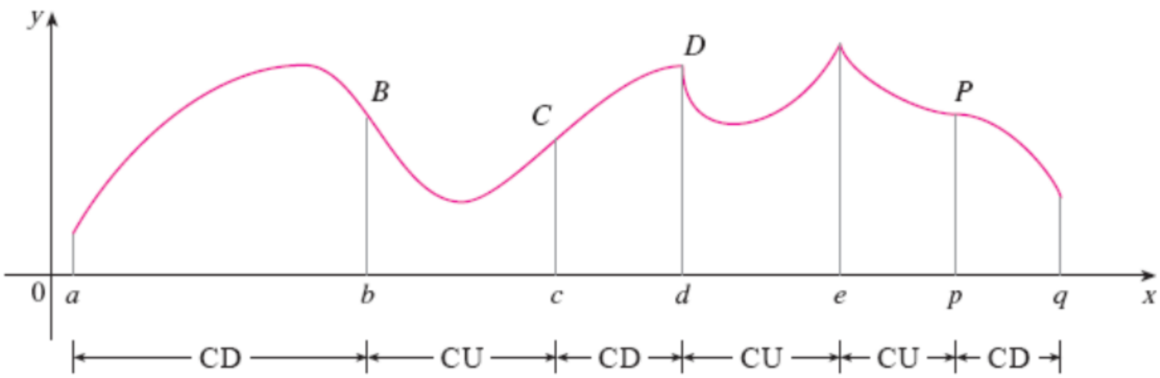
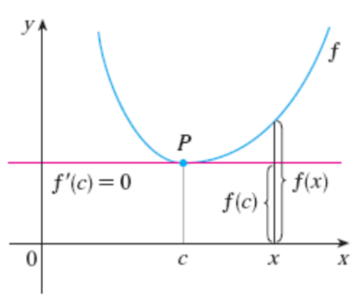


Figure 3.1.2: Peicewise concave upwards and downwards

Concavity Test

(a) If $f''(x) > 0$ for all x in I, then the graph of f is concave upward on I.

(b) If $f''(x) < 0$ for all x in I, then the graph of f is concave downward on I.



Suppose c is a critical point of f. Then we have:

The Second Derivative Test Suppose f'' is continuous near c.

(a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c.

(b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c.

Based on the above facts, we are now going to prove part (a) of the second derivative test.

Proof of part (a) of the second derivative test: In a neighborhood of the point (a, b), we are going to consider a disk. Through (a, b), we are going to consider a particular direction, say the direction of an arbitrarily fixed unit vector $\hat{u}$. Consider a vertical plane through (a, b) in the direction of $\hat{u}$. This plane will pass through the point $(a, b, f(a, b))$. This plane is going to cut the surface $z = f(x, y)$ in a curve.

We are going to prove that for this curve, the function has a minimum at the point (a, b). Now, $\hat{u}$ was an arbitrary direction. So at the point (a, b), we can consider any arbitrary direction, and for all the directions, we are going to arrive at the same conclusion.

We have to consider the second order directional derivative of f at (a, b) along the direction $\hat{u}$, and prove that it is > 0. For a function of one variable, we would have taken the second order ordinary derivative and proved that it is > 0. Here since we are considering an arbitrary direction, we must prove that $D^2_{\hat{u}} f(a, b) > 0$.

Let $\hat{u} =< h, k >$. Then

$$D_{\hat{u}} f(x, y) = f_x h + f_y k.$$

And

$$\begin{aligned} D^2_{\hat{u}} f(x, y) &= D_{\hat{u}}(D_{\hat{u}} f(x, y)) = D_{\hat{u}}(f_x h + f_y k) \\ &= (f_x h + f_y k)_x h + (f_x h + f_y k)_y k \\ &= f_{xx} h^2 + 2 f_{xy} hk + f_{yy} k^2. \end{aligned}$$

Therefore,

$$D^2_{\hat{u}} f(a, b) = f_{xx}(a, b) h^2 + 2 f_{xy}(a, b) hk + f_{yy}(a, b) k^2$$

$$= f_{xx}\left(h^2 + 2h\frac{f_{xy}}{f_{xx}}k + \frac{f_{xy}^2}{f_{xx}^2}k^2\right) + f_{yy}k^2 - \frac{f_{xy}^2}{f_{xx}}k^2 \text{ evaluated at } (a,b)$$

$$= f_{xx}\left(h + \frac{f_{xy}}{f_{xx}}k\right)^2 + \frac{k^2}{f_{xx}}\left(f_{xx}f_{yy} - f_{xy}^2\right) \text{ evaluated at } (a,b)$$

Recall that our assumptions were $D(a,b) > 0$ and $f_{xx}(a,b) > 0$. It hence follows that the above expression is > 0. That is,

$$D_{\hat{u}}^2 f(a,b) > 0.$$

Now since $D = f_{xx}f_{yy} - f_{xy}^2$ is the subtraction of the products of some continuous functions, it follows that D is continuous at (a,b). Also, f_{xx} is continuous at (a,b). So $D_{\hat{u}}^2 f$ is definitely continuous at (a,b). That means if we consider a small disk $D_\delta(a,b)$ around (a,b), then in that disk, we will always have $D_{\hat{u}}^2 f > 0$. Now if we consider any arbitrary direction at the point (a,b), then we are going to arrive at the same conclusion. So in a neighborhood of (a,b), the surface is going to be concave upwards. This implies that f has a local minimum at (a,b). $\square$

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